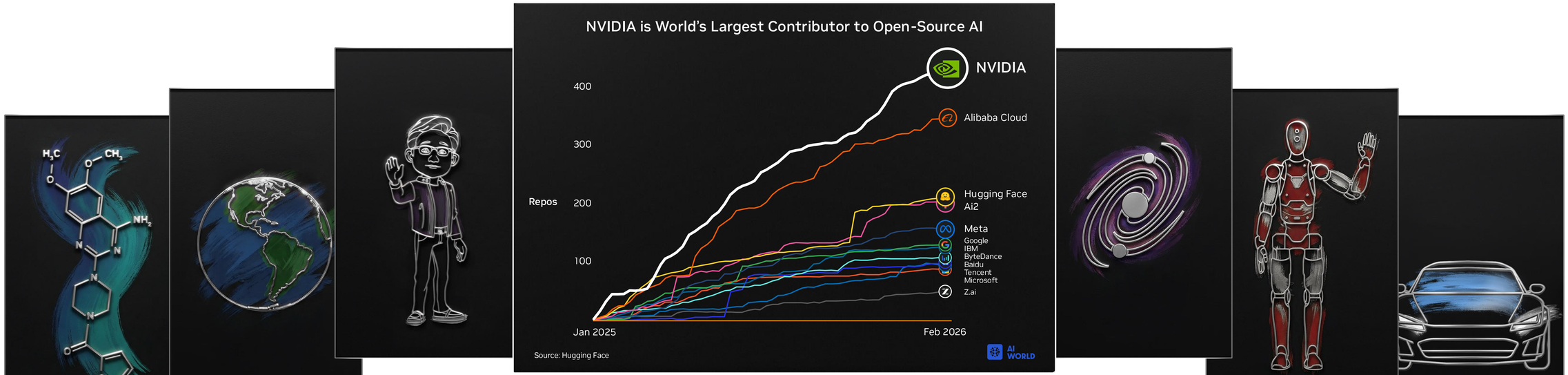
MICCAI Announcements

MICCAI 2026

NVIDIA Leading Open Model Ecosystem



BioNeMo
Biology AI

Earth-2
Physics

Nemotron
Agentic AI

Cosmos
Physical AI

GR00T
Robotics

Alpamayo
Autonomous

Artificial Analysis Intelligence Index

Artificial Analysis Output

VLM3D

ViDoRe, MTEB, MMTEB

OpenASR

Physical Reasoning MVPBench, IntPHys, CasualVQA

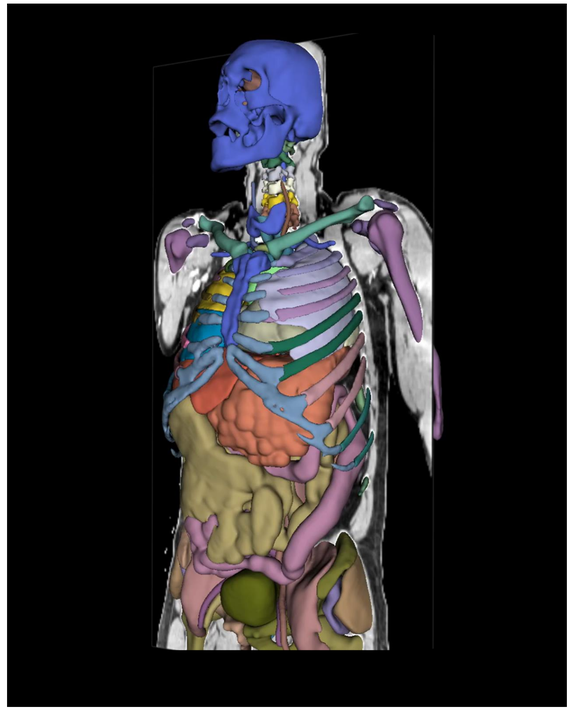
OCRBench

PAIBench

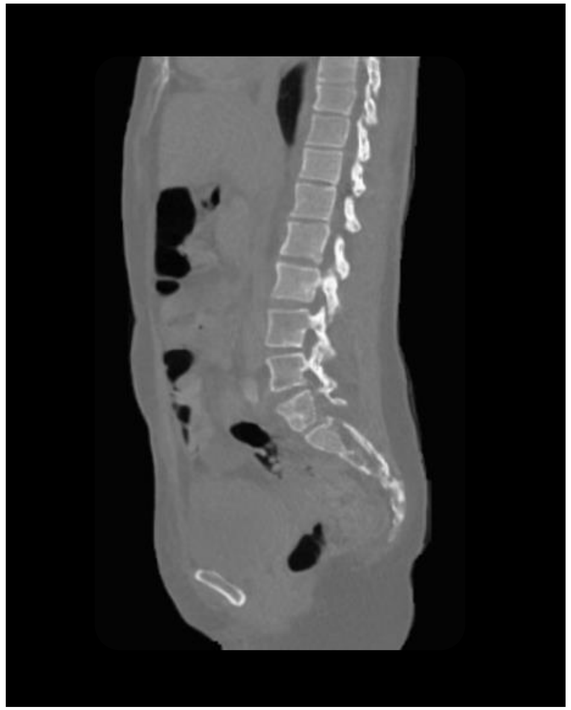
NVIDIA

NVIDIA Medical AI Open Models

Multimodal foundation models accelerating clinical innovation



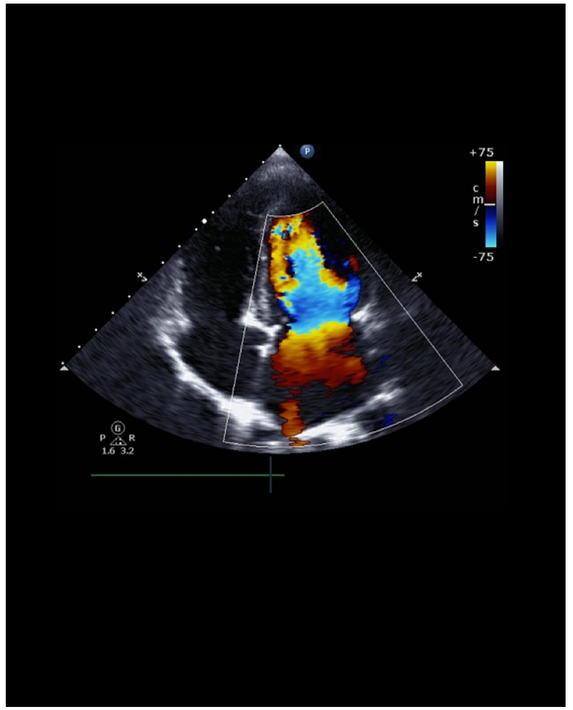
Segment
3D automatic and interactive annotation



Generate
3D synthetic dataset generation



Reason
Chain of thought radiology reasoning



Raw2Insights
Physics-based AI image reconstruction

VLM3D

CMRxRecon2025

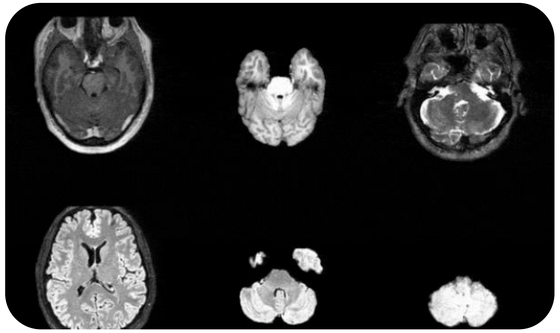
Announcing NVIDIA Medical AI Releases

Open models, simulation tools, and open data initiatives for the medical imaging research community



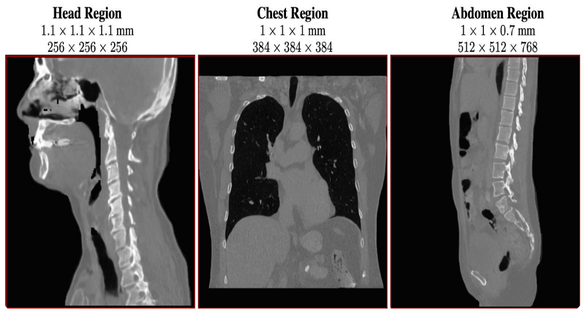
MONAI 1.6

A focused MONAI Core update with HyenaNDUNETR for segmentation research, richer evaluation tools, and stronger workflow and data reliability.



NV-REASON-CT

NV-Reason-CT delivers AI-powered chain-of-thought reasoning for volumetric CT imaging, providing transparent, step-by-step radiology insights grounded in 3D image evidence.



Open-H-4D Data Initiative

A community initiative to expand commercially usable 4D medical imaging data for patient-specific physiological simulation and AI-solver research.

Finetune it *nv_segment_ct_finetune*

```
$ npx skills add NVIDIA-Medtech/medical-AI-skills
  --skill nv-segment-ct-finetune
```

Try saying

> *"I have an MSD-style CT dataset at /data/msd_case. Do the shortest smoke-scale finetune check first."*

Agentic Skills for Medical AI

A verified catalog of agent-callable skills for medical imaging: DICOM utilities, CT and MR segmentation, synthetic data generation, and image reasoning, all built on MONAI workflows and NVIDIA MedTech models.

NVIDIA

Research Activation (MICCAI)

MICCAI recognition and results

- **CancerVerse:** Best Paper and Young Scientist Award nominee. Includes 24,168 CT scans across 13 cancer types, with a nine-point sensitivity gain at 0.5 false positives per scan (1M+ downloads). Paper, Dataset
- **Photon-Counting CT / SUMI:** Selected for a MICCAI Spotlight. SUMI uses photon-counting CT as a reference for enhancing conventional CT, with radiologists assessing clinical realism. Paper, Related OpenVAE GitHub, Model weights
- **MIAgent:** Agent harness for long-running medical imaging research. Placed second in PENGWIN Task 3 and third in Tasks 1 and 2. PENGWIN results

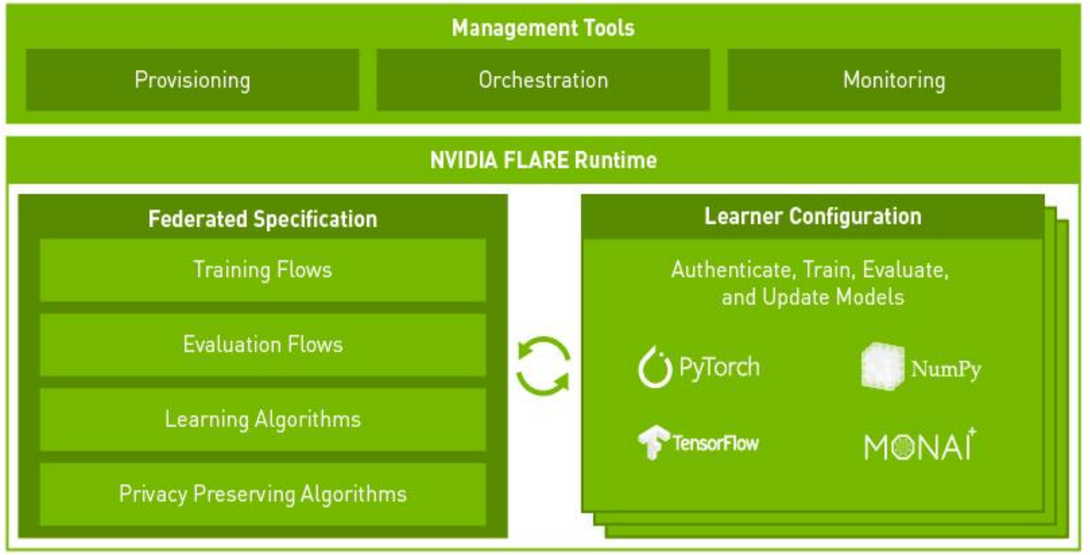
Open research releases

- **Cosmos-H-Dreams:** Real-time, action-conditioned generative simulation for surgical robotics. Paper, GitHub, Model
- **AutoMedBench:** Benchmark for medical AI agents with 96 task-tier combinations across seven research tracks. Benchmark, Paper, GitHub
- **MONAI Physio:** Toolkit for creating personalized physiological digital twins from 3D and 4D medical images. Paper planned. GitHub, Documentation
- **Cosmos-H-Surgical:** Cosmos3-Nano surgical video model combining video prediction and control-conditioned transfer in one checkpoint. Paper, GitHub, Model
- **FedCoRe:** Healthcare Federated Learning with Missing Modalities in NVFlare. Paper, GitHub

NVIDIA

NVIDIA FLARE OSS SDK

- **Enterprise-Scale & Production-Ready:** Mature, secure, and scalable.
- **Privacy & Compliance:** differential privacy, homomorphic encryption, and confidential computing.
- **Framework & Model Agnostic:** Supports any ML framework or model type.
- **Simple API:** APIs that federate existing training code in under 10 lines.
- **End-to-End Lifecycle:** From simulation to large-scale production.
- **Flexible Deployment:** Supports on-prem, cloud, and hybrid environments.
- **Federated Compute:** Federated statistics, analytics, evaluation, and data process — not just model training.



https://developer.nvidia.com/flare

Get started

NVIDIA FLARE Major Customers

Healthcare & life science

Roche
Lilly
MAYO CLINIC
MOFFITT CANCER CENTER
FATE FEDERATED AI THERAPEUTIC ENGINEERING
AbbVie • Amgen • AstraZeneca • Johnson & Johnson • UCB
CAIA

AI Factories & National AI Centers

AI2F
GENCI HPC at the service of knowledge
BriCS Bristol Centre for Supercomputing
JÜLICH Forschungszentrum | JÜLICH SUPERCOMPUTING CENTRE

Transportation, Wireless etc.

NVIDIA

AV model training

Earth-2 Data Foundation (Federated Data Mesh)

RAN – Wireless

Holoscan – Fed Stats

Government Agencies

Sandia National Laboratories
OAK RIDGE National Laboratory
衛生福利部 Ministry of Health and Welfare
Los Alamos NATIONAL LABORATORY EST.1943
Lawrence Livermore National Laboratory

Financial Services

RBC
Swift
J.P.Morgan
NPCI (National Payments Corporation of India)
Node16.ai
BNY

ISV and GSI

Rhino Federated Computing
Duality
APHERIS
Deloitte.
oasys now
FLIP
廣達電腦 Quanta Computer

NVFlare Days Sep ’26 https://events.nvidia.com/flare-day-2026

Auto-FL: AI Agents for Federated Learning

Automating the Search for Better FL Solutions



- Define project goal
- Setup datasets
- Provide compute budget
- Guide agents via prompt

AI Agents



- Explore FL recipes & parameters
- Compare results
- Literature search & review
- Produce best FL solution

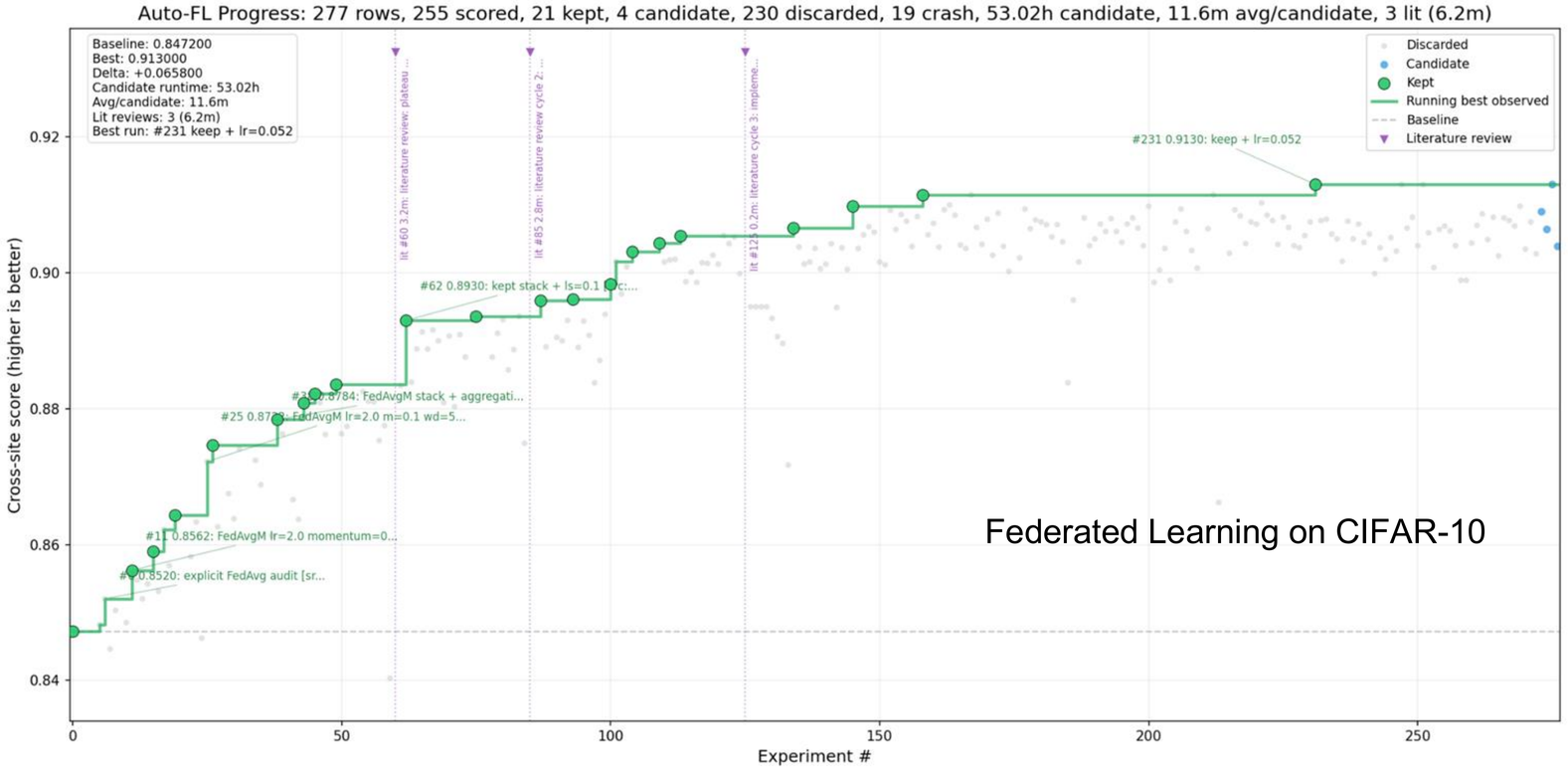
Python Program

NVFlare Environment

runs federated experiments

Models

Dataset 1 | Dataset 2 | Dataset 3 | Dataset …



Try Auto-FL

GitHub: github.com/NVIDIA/NVFlare/tree/main/research/auto-fl-research
Webinar: youtu.be/-8kqkMUZs2c

NVIDIA-Sponsored Workshops at MICCAI 2026

14 sessions across medical imaging, digital twins, surgical AI, federated learning, and data engineering

- **RIME: Reconstruction & Imaging Motion Estimation**
- **DT4H: Digital Twin for Health**
- **COLAS: Collaborative Intelligence in Image-guided Surgery**
- **GRAIL: GRaphs in biomedicAI Image anaLysis**
- **ShapeMI: Shape in Medical Imaging**
- **DeCaF: Distributed, Collaborative & Federated Learning**
- **ASMUS: Advances in Simplifying Medical Ultrasound**
- **SafeSurg: AI for Safe Surgery**
- **OMIA13: Multimodal AI in Ophthalmic Surgery**
- **HAIC: Human-AI Collaborative Systems**
- **Off-Grid: Grid-Free Methods in Medical Imaging**
- **DA-OCT: Domain Adaptation for Retinal OCT**
- **EndoVis Challenge: Endoscopic Vision**
- **DEMI: Data Engineering in Medical Imaging**

NVIDIA

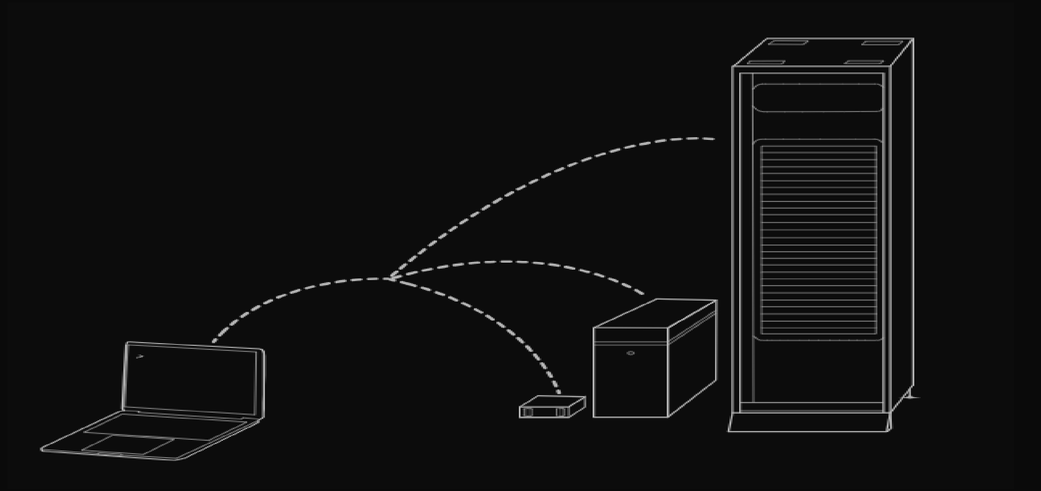
DeCaF Workshop Awards

NVIDIA Brev credits + DGX Spark

Terms and Conditions Apply

BUILD IN THE CLOUD

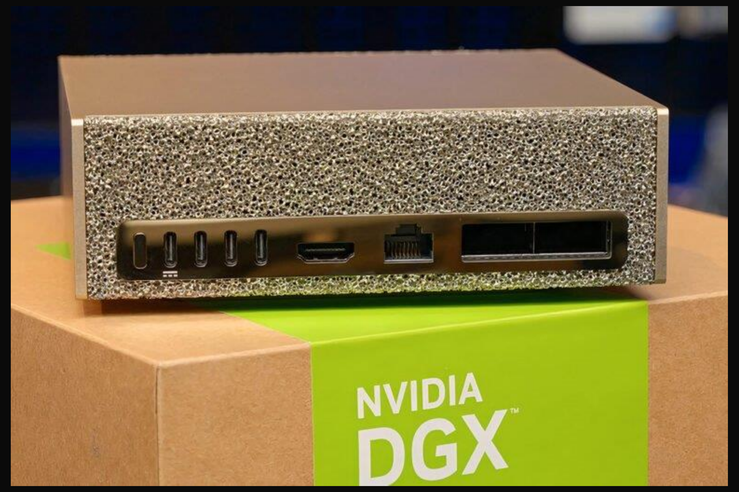
$350 NVIDIA Brev Credits



Fast access to preconfigured GPU instances across cloud providers, so researchers can build, train, test, and run AI workloads without managing infrastructure.

DESKTOP AI SUPERCOMPUTER

DGX Spark



A compact Grace Blackwell desktop AI system with 128 GB of unified memory, built for developing, fine-tuning, and running advanced AI models locally.

One award package will be selected for each workshop.


NVIDIA

MICCAI 2027 CHALLENGE · ANNOUNCEMENT



FeTS++ 2027

Federated Tumor Segmentation and Classification Challenge

One federated-learning dataset, two tasks. Participants develop aggregation strategies and client training behavior — patient data never leaves its institution.

FeTS 2021 · FeTS 2022 · FeTS 2024 · **FeTS++ 2027**

+ classification + NVFLARE

MICCAI · NIH National Cancer Institute · NVIDIA · Synapse Challenges · Powered by Sage Bionetworks · IU

FETS++ 2027 · MICCAI CHALLENGE

What's New in FeTS++

01 Framework transition — NVFLARE

Infrastructure migrates to NVIDIA NVFLARE. Participants customize client training behavior and the server-side aggregator —

Targeting generalizable FL aggregatin method

02 A larger, more diverse federated dataset

Multi-institutional multi-parametric MRI from the BraTS 2025 Lighthouse Challenge, now spanning multiple tumor entities beyond glioblastoma — each case assigned to a de-identified institution for realistic cross-site federated simulation.

3,500+

patients · multi-parametric MRI

03 Two official tasks under one dataset

TASK 1 · FEDERATED SEGMENTATION (Opens Today!)

Aggregate segmentation models trained at individual institutions — aggregation, client selection, local training per round, training efficiency.

TASK 2 · FEDERATED CLASSIFICATION (Opens on Nov 1st)

Distinguish glioma from meningioma on the same training data.

Targeting challenge from real-world institutional partition with label skewing.

Thanks to NVIDIA! We are providing $250 in Brev cloud credits for registered team (one coupon per team, limited availability.)



Register Today!